

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. Computer Engineering Sem - II University Examination, 2010-11

CE 713 Data Mining and Business Intelligence (DM&BI)

Date: 19.05.11, Thursday Time: 10:00 a.m to 01:00 p.m Maximum Marks: 70

- Instructions:** (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work to be done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

1. What are the challenges in the various fields associated with Data Mining? 3
2. What do you mean by ETL with respect to DWDM? 3
3. What are the things necessary for maintaining a data warehouse? 3
4. How are multiple facts handled? What are the schemas used for handling a single fact and for handling (using) multiple facts? 2

Q-2

- [A] Suppose that a warehouse consists of 4 dimensions date, viewer, cinema hall and movie and two measures count and charge (ticket fees). The viewer can be child < 5, above 5, adults or senior with each category having its own charge rate. 4

1. Draw the star schema model.
2. What specific OLAP operations one should perform in order to list the total charge paid by adults at the cinema hall "Rajhans" in 2010?

- [B] What are the stages of processing a relational query? How to improve query processing? Briefly explain the query processing strategies to be adopted for efficient retrieval of the data from the warehouse? 4

OR

- [B] Concurrency control and recovery mechanisms are not applied for OLAP. Justify. 4

- [C] What is aggregation? Is it really necessary? If yes, why detail data in warehouse is necessary? If no, justify. 4

OR

- [C] How do optimization techniques apply to MOLAP? Explain key based addressing of ROLAP. 4

Q-3

- [A] Give the significance and importance of the Terms: Mean, Variance, Standard deviation and central tendency. 4

OR

- [A] Construct a data cube from table. Is this a dense or sparse data cube? If it is sparse, identify the cells that are empty. 4

Product_id	Loc_id	No. sold.
1	1	10
1	3	6
2	1	5
2	2	22

- [B] Substantiate your answer for the usage of the terms: OLAP, OODBMS, Data cubing and normalization. 4

OR

- [B] Data cleaning routines can be used to fill in missing values, smooth noisy data, identify outliers and correct the data inconsistency. Explain how to deal with each of them in terms of problems, in brief. 4
- [C] Describe briefly the rules which are useful in setting up a reliable Data Mining. 4

OR

- [C] Explain the relationship between fact table and dimension table by explaining functionalities of both. 4

SECTION-II

Q-4

1. For what purpose statistics and machine learning are used in data mining? 3
2. How do we find the interestingness of association rules? For what purpose association rules are being used? 4
3. Compare and contrast: Data Cube Approach and AOI Approach. 4

Q-5

- [A] Suppose that you are employed as a Data Mining consultant for an Internet Search Engine Company. Describe how Data Mining can help the company by giving specific examples of how techniques, such as clustering, classification, association rule mining and anomaly detection can be applied. 4
- [B] Explain patterns and come up with the most elementary kinds of patterns from which all other kinds of patterns can be derived. 4
- [C] What is the relationship between data compression and data mining? 4

OR

Q-5

- [A] How can frequent item set mining be used for classification? What are different associations with two phases of association rule mining? 4
- [B] What are precision and recall? How do they differ from accuracy? 4
- [C] Following is the series of data for the price: 4
80,40,210,150,210,240,340,280,210,390.
 1. Partition the data into equi-depth bins
 2. Smooth the data by bin means and find outliers, if any.

Q-6

- [A] Differentiate between: Bayesian Classifier and Neural Network Classifier. 4
- [B] Generate the frequent-3 itemsets with candidate generation for the following transaction database. 4

T1	I1,I2,I5
T2	I2,I4
T3	I1,I2,I4
T4	I1,I3
T5	I2,I3
T6	I1,I3
T7	I1,I2,I3,I5
T8	I1,I2,I3

- [C] Define the problems in terms of Clustering? How does it differ from classification? Substantiate your answer. 4

OR

- [C] Here with given a Purchase data of games and Videos. Out of 10,000 transactions analyzed, 6000 customer transactions included games, 7500 transactions included videos and 4000 included both games and videos. Prepare a contingency table and check which type of correlation exists between games and videos. 4